

CS5187 Vision and Image - Exam Solution

Detailed Solutions & Explanations

Semester B 2019/20

Section A: Short Question (50%)

Q1

Question: What are the parameters required to set when generating a pyramid of Gaussian images for a picture? Why pyramid of Gaussian is applied in various image processing algorithms?

Translation: 在为图片生成高斯金字塔 (Pyramid of Gaussian) 时, 需要设置哪些参数? 为什么高斯金字塔被应用于各种图像处理算法中?

Answer:

- **Parameters:**

1. The number of levels (scales) in the pyramid.
2. The subsampling rate (usually 2, meaning the image size is halved at each level).
3. The smoothing parameter σ (standard deviation) for the Gaussian kernel used before subsampling.

- **Reason for Application:**

1. **Scale Invariance:** It allows algorithms to detect features (like edges, corners, or objects) at multiple scales/sizes.
2. **Computational Efficiency:** Algorithms can perform coarse processing on smaller images (higher levels) to get a rough estimate, and then refine the result on larger images (lower levels), which is much faster than processing only the high-resolution image.

解答 (中文):

- **所需参数:**

1. 金字塔的层数 (Scales) 。
2. 下采样率 (通常为2, 意味着每层的图像尺寸减半) 。
3. 下采样前使用的高斯核的平滑参数 σ (标准差) 。

- **应用原因:**

1. **尺度不变性:** 它允许算法在多个尺度/大小上检测特征 (如边缘、角点或物体) 。
2. **计算效率:** 算法可以在较小的图像 (高层) 上进行粗略处理以获得初步估计, 然后在较大的图像 (低层) 上细化结果。这种 “从粗到细” 的策略比仅处理高分辨率图像快得多。

Q2

Question: Calculate the normalized co-occurrence matrix for the given image patch below using the vector $d = (1, 2)$ which specifies the spatial relation between two pixels. What is the property of this image patch? Note: $d = (dr, dc)$ is the relation in row (dr) and column (dc).

```
2 0 0 0 0
1 0 2 0 0
1 0 1 0 2
0 0 1 0 1
0 0 0 0 1
```

Translation: 使用向量 $d = (1, 2)$ 计算下面给出的图像块的归一化共生矩阵 (Normalized Co-occurrence Matrix)，该向量指定了两个像素之间的空间关系。该图像块有什么性质？注意： $d = (dr, dc)$ 表示行 (dr) 和列 (dc) 的关系。

Answer: The vector $d = (1, 2)$ means we look at pixel pairs $(I(r, c), I(r + 1, c + 2))$. The valid pairs in the 5×5 image are:

- $(0,0) \rightarrow (1,2)$: $2 \rightarrow 2$
- $(0,1) \rightarrow (1,3)$: $0 \rightarrow 0$
- $(0,2) \rightarrow (1,4)$: $0 \rightarrow 0$
- $(1,0) \rightarrow (2,2)$: $1 \rightarrow 1$
- $(1,1) \rightarrow (2,3)$: $0 \rightarrow 0$
- $(1,2) \rightarrow (2,4)$: $2 \rightarrow 2$
- $(2,0) \rightarrow (3,2)$: $1 \rightarrow 1$
- $(2,1) \rightarrow (3,3)$: $0 \rightarrow 0$
- $(2,2) \rightarrow (3,4)$: $1 \rightarrow 1$
- $(3,0) \rightarrow (4,2)$: $0 \rightarrow 0$
- $(3,1) \rightarrow (4,3)$: $0 \rightarrow 0$
- $(3,2) \rightarrow (4,4)$: $1 \rightarrow 1$

Total pairs = 12. Counts: $(0, 0) : 6, (1, 1) : 4, (2, 2) : 2$. Normalized Matrix (divide by 12):

$$P = \begin{pmatrix} 6/12 & 0 & 0 \\ 0 & 4/12 & 0 \\ 0 & 0 & 2/12 \end{pmatrix} = \begin{pmatrix} 0.5 & 0 & 0 \\ 0 & 0.33 & 0 \\ 0 & 0 & 0.17 \end{pmatrix}$$

Property: The matrix is diagonal. This implies a strong structural correlation or periodicity in the direction $(1, 2)$. The pixel value at any location (r, c) perfectly predicts the value at $(r + 1, c + 2)$.

解答 (中文): 向量 $d = (1, 2)$ 意味着我们检查像素对 $(I(r, c), I(r + 1, c + 2))$ 。统计结果显示： $(0, 0)$ 出现6次， $(1, 1)$ 出现4次， $(2, 2)$ 出现2次，其余组合为0。归一化矩阵如上所示。**性质:** 矩阵是对角矩阵。这意味着图像在方向 $(1, 2)$ 上具有很强的结构相关性或周期性。任何位置 (r, c) 的像素值完全等于位置 $(r + 1, c + 2)$ 的像素值。

Q3

Question: The Harris operator detects corner by computing the two eigenvalues λ_1 and λ_2 of the Harris matrix, where $\lambda_1 > \lambda_2$. A threshold, T , is then set to determine whether a corner is detected. Which of the following criterion is more appropriate to detect corner? Justify your answer.

(i) $\lambda_1 > T$ (ii) $\lambda_1 \lambda_2 > T$

Translation: Harris 算子通过计算 Harris 矩阵的两个特征值 λ_1 和 λ_2 来检测角点，其中 $\lambda_1 > \lambda_2$ 。然后设定阈值 T 来确定是否检测到角点。以下哪种标准更适合检测角点？证明你的答案。(i) $\lambda_1 > T$ (ii) $\lambda_1 \lambda_2 > T$

Answer: Criterion (ii) $\lambda_1 \lambda_2 > T$ is more appropriate. Justification: A corner is characterized by significant intensity changes in all directions. In terms of eigenvalues of the structure tensor (Harris matrix):

- If both λ_1 and λ_2 are small, it is a flat region.
- If λ_1 is large and λ_2 is small (so $\lambda_1 > T$ but $\lambda_1 \lambda_2$ is small), it represents an **edge**, not a corner.
- If both λ_1 and λ_2 are large, it represents a **corner**.

Criterion (i) $\lambda_1 > T$ would incorrectly classify edges as corners because an edge has one large eigenvalue. Criterion (ii) $\lambda_1 \lambda_2 > T$ (which is the determinant of the matrix) ensures that **both** eigenvalues are sufficiently large, which is the definition of a corner.

解答 (中文): 标准 (ii) $\lambda_1 \lambda_2 > T$ 更合适。理由: 角点的特征是在所有方向上都有显著强度变化。就结构张量 (Harris 矩阵) 的特征值而言:

- 如果 λ_1 和 λ_2 都很小，则是平坦区域。
- 如果 λ_1 很大而 λ_2 很小 (满足 $\lambda_1 > T$ 但乘积很小)，这代表一条**边缘**，而不是角点。
- 只有当 λ_1 和 λ_2 都很大时，才代表一个**角点**。

标准 (i) 会错误地将边缘归类为角点。标准 (ii) (即矩阵的行列式) 确保了**两个**特征值都足够大，符合角点的定义。

Q4

Question: A set of 100 pixels are matched between the two images captured by a pair of uncalibrated cameras. It is estimated that at least 70% of matchings are correct. To estimate the disparity map (implies Fundamental Matrix estimation here) for the two images, RANSAC algorithm is applied. What is the minimum number of iterations required by RANSAC to estimate correct answer with 99% probability? Show the steps of calculation.

Translation: 一对未校准的相机拍摄的两张图像之间匹配了 100 个像素点。估计至少 70% 的匹配是正确的。应用 RANSAC 算法来估计两张图像的视差图 (此处隐含指估计基础矩阵)。RANSAC 至少需要多少次迭代才能以 99% 的概率估计出正确答案？展示计算步骤。

Answer: Assumptions: Since the cameras are uncalibrated and we are relating two images, we typically estimate the **Fundamental Matrix**, which requires a minimum sample size of $s = 8$ (using the 8-point algorithm) or $s = 7$. We assume $s = 8$ for a standard linear solution.

- Probability of success required, $p = 0.99$.
- Proportion of inliers (correct matches), $w = 0.70$.
- Minimum sample size, $s = 8$.

Calculation: The formula for the number of iterations N is:

$$N = \frac{\log(1-p)}{\log(1-w^s)}$$

1. Calculate probability of picking 8 inliers in one try: $w^s = 0.7^8 \approx 0.0576$. 2. Calculate probability of failure in one try: $1 - w^s = 1 - 0.0576 = 0.9424$. 3. Apply formula:

$$N = \frac{\log(1-0.99)}{\log(0.9424)} = \frac{\log(0.01)}{\log(0.9424)} \approx \frac{-2}{-0.02576} \approx 77.6$$

Result: We round up to the nearest integer. Minimum iterations $N = 78$. (Note: If assuming $s = 7$, $N \approx 54$. If assuming homography $s = 4$, $N \approx 17$. Given "uncalibrated cameras", $s = 8$ is the safest standard assumption).

解答 (中文): 假设: 由于相机未校准且涉及两张图像的关系, 通常需要估计**基础矩阵 (Fundamental Matrix)**。这通常需要最少 $s = 8$ 个点 (8点算法)。

- 所需成功概率 $p = 0.99$ 。
- 内点 (正确匹配) 比例 $w = 0.70$ 。
- 最小样本数 $s = 8$ 。

计算步骤: 迭代次数公式为: $N = \frac{\log(1-p)}{\log(1-w^s)}$ 。 1. 一次抽取全为内点的概率: $0.7^8 \approx 0.0576$ 。 2. 一次失败的概率: $1 - 0.0576 = 0.9424$ 。 3. 计算 N : $N = \frac{\log(0.01)}{\log(0.9424)} \approx 77.6$ 。 **结果:** 向上取整, 最少需要 ****78**** 次迭代。

Q5

Question: A photographer takes 12 pictures using a perspective projection camera mounted on a tripod. The tripod is rotated by 30 degree in angle each time before taking a picture. Can the scene depth be recovered using the pictures taken in 360 degree? Explain your answer.

Translation: 摄影师使用安装在三脚架上的透视投影相机拍摄了 12 张照片。每次拍摄前, 三脚架旋转 30 度。能否利用这 360 度拍摄的照片恢复场景深度? 解释你的答案。

Answer: No, the scene depth cannot be recovered. Explanation: Depth recovery (structure) from multiple images requires **parallax**, which is produced by the translation of the camera center (baseline). In this scenario, the camera is only **rotated** around its optical center on a tripod; there is zero translation. Without translation, all light rays pass through the same optical center, forming a panorama (mosaic) rather than a stereo setup. Geometrically, it is impossible to triangulate points in 3D space without a baseline.

解答 (中文): 不能, 无法恢复场景深度。解释: 从多张图像恢复深度 (结构) 需要**视差 (parallax)**, 而视差是由相机中心的**平移**产生的。在这个场景中, 相机仅围绕其光心在三脚架上**旋转**, 没有发生平移。如果没有平移, 所有光线都穿过同一个光心, 这会形成全景图 (Panorama), 而不是立体视觉设置。在几何上, 没有基线 (baseline) 就不可能对 3D 空间中的点进行三角测量。

Q6

Question: Consider a set of parallel lines on the X-Z plane, where Z is the optical axis of a camera. Why these lines will intersect when being projected on the image plane (X-Y plane)? What is the property of the intersection point on the image?

Translation: 考虑 X-Z 平面上的一组平行线，其中 Z 是相机的光轴。为什么这些线投影到图像平面 (X-Y 平面) 上时会相交？图像上的交点有什么性质？

Answer: Why intersect: In perspective projection, parallel lines in 3D world space (that are not parallel to the image plane) appear to converge to a single point in the 2D image. This is because objects further away appear smaller (scaling by $1/Z$). As $Z \rightarrow \infty$, the projection of the lines converges. **Property:** The intersection point is called the **Vanishing Point**. Its property is that it corresponds to the projection of the **direction vector** of the parallel lines at infinity. For lines on the X-Z plane parallel to the Z-axis, the vanishing point will be at the principal point (image center).

解答 (中文): 为什么相交: 在透视投影中，3D 空间中的平行线（如果不平行于图像平面）在 2D 图像中看起来会汇聚到一点。这是因为远处的物体看起来更小（按 $1/Z$ 缩放）。当 Z 趋向无穷大时，这些线的投影就会汇聚。**性质:** 该交点称为**消失点 (Vanishing Point)**。其性质是：它对应于这些平行线在无穷远处的**方向向量**在图像上的投影。

Q7

Question: Economic beehoon... Suppose you want to classify, segment and count the food items in the breakfast (see the sample image). Which one of the following neural networks you will use: faster R-CNN, fully convolutional network (FCN), Mask R-CNN? Explain your choice.

Translation: 经济米粉... 假设你想对早餐中的食物项目进行分类、分割和计数（见示例图）。你会使用以下哪种神经网络：Faster R-CNN，全卷积网络 (FCN)，Mask R-CNN？解释你的选择。

Answer: Choice: Mask R-CNN. Explanation: The task requires three things: **classify**, **segment**, and **count** individual items (e.g., distinguish between two different sausages).

- **Faster R-CNN** is good for detection (bounding box) and classification, but it does not provide pixel-level segmentation.
- **FCN** performs semantic segmentation (classifying pixels), but it cannot distinguish between different instances of the same class (e.g., two touching sausages would be merged into one blob), making counting difficult.
- **Mask R-CNN** is an **Instance Segmentation** network. It extends Faster R-CNN by adding a branch for predicting segmentation masks on each Region of Interest (RoI). It successfully separates instances (good for counting), classifies them, and generates pixel-wise masks (segmentation).

解答 (中文): 选择: Mask R-CNN。解释: 任务要求三件事：**分类**、**分割**和**计数**个体项目（例如，区分两根不同的香肠）。

- **Faster R-CNN** 擅长检测（边界框）和分类，但不提供像素级的分割。
 - **FCN** 执行语义分割（分类像素），但它无法区分同一类别的不同实例（例如，两根接触的香肠会合并成一块），这使得计数变得困难。
 - **Mask R-CNN** 是一个**实例分割**网络。它通过在每个感兴趣区域 (RoI) 上添加预测分割掩码的分支来扩展 Faster R-CNN。它能成功分离实例（利于计数），对其进行分类，并生成像素级掩码（分割）。
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Q8

Question: Why structure-from-motion is considered as a chicken-and-egg problem? Suppose the inputs include 20 images and 2000 scene points, how many unknown variables are required to be solved for structure-from-motion?

Translation: 为什么运动恢复结构 (SfM) 被认为是“鸡生蛋，蛋生鸡”的问题？假设输入包括 20 张图像和 2000 个场景点，SfM 需要求解多少个未知变量？

Answer: Chicken-and-egg problem: To determine the 3D structure of the scene points, we need to know the camera poses (motion). However, to determine the camera poses, we need to know the correspondences of 3D structure points. Neither is known initially, so we must estimate them simultaneously (usually via Bundle Adjustment). **Unknown variables:**

- **Motion (Cameras):** Each camera has 6 degrees of freedom (3 rotation, 3 translation). For 20 images: $20 \times 6 = 120$ variables.
- **Structure (Points):** Each scene point has 3 coordinates (X, Y, Z). For 2000 points: $2000 \times 3 = 6000$ variables.

Total unknowns: $6000 + 120 = 6120$ variables. (Note: The actual solvable degrees of freedom is slightly less due to gauge freedom—scale, rotation, and translation of the whole world frame are arbitrary—but the number of raw unknown parameters is 6120).

解答 (中文): 鸡与蛋的问题: 为了确定场景点的 3D 结构，我们需要知道相机姿态（运动）。然而，为了确定相机姿态，我们需要知道 3D 结构点的对应关系。两者最初都是未知的，因此必须同时估计它们。 **未知变量数量:**

- **运动 (相机):** 每个相机有 6 个自由度（3 旋转，3 平移）。20 张图像： $20 \times 6 = 120$ 个变量。
- **结构 (点):** 每个场景点有 3 个坐标 (X, Y, Z)。2000 个点： $2000 \times 3 = 6000$ 个变量。

总未知数: $6000 + 120 = 6120$ 个变量。

Q9

Question: Why stereo vision and optical flow estimation are possible to be solved by the same algorithm? Briefly describe how convolutional neural network can be modified for both problems of stereo vision and optical flow estimation.

Translation: 为什么立体视觉和光流估计可以通过相同的算法解决？简要描述如何修改卷积神经网络以同时适用于立体视觉和光流估计问题。

Answer: Reason: Both problems are fundamentally about finding **correspondences** between two images. Stereo vision finds the displacement (disparity) of pixels along horizontal lines (1D search) between a left and right image. Optical flow finds the displacement vector (u, v) (2D search) between two time frames t and $t + 1$. **CNN Modification:**

1. **Input:** The network accepts a pair of images stacked together (channel-wise) or processed via a Siamese network to extract features which are then correlated (Correlation Layer).
2. **Output Layer:**
 - For **Stereo:** The output layer has 1 channel (scalar value per pixel) representing disparity d .
 - For **Optical Flow:** The output layer has 2 channels representing the vector (u, v) .

The internal architecture (encoder-decoder, feature matching) remains largely similar.

解答 (中文): 原因: 这两个问题本质上都是为了在两张图像之间寻找**对应关系**。立体视觉寻找左右图像之间沿水平线的像素位移 (视差, 1D 搜索)。光流寻找时间帧 t 和 $t + 1$ 之间的位移向量 (u, v) (2D 搜索)。**CNN 修改:**

1. **输入:** 网络接收堆叠的一对图像, 或通过孪生网络提取特征, 然后计算相关性 (Correlation Layer)。
 2. **输出层:**
 - 对于**立体视觉**: 输出层有 1 个通道, 代表视差 d 。
 - 对于**光流**: 输出层有 2 个通道, 代表向量 (u, v) 。
- 内部架构 (编码器-解码器, 特征匹配) 基本保持相似。

Q10

Question: What does "receptive field" mean in video processing? Describe how does receptive field is being used in 3D convolutional neural network for activity classification.

Translation: 在视频处理中, “感受野 (receptive field)” 是什么意思? 描述感受野如何在用于活动分类的 3D 卷积神经网络中使用。

Answer: Meaning: In video processing, the receptive field refers to the region in the input video volume (spatial height/width $\times$ temporal frames) that influences a specific neuron's activation in a deeper layer. It is **spatio-temporal**, covering a specific patch of pixels over a specific duration of time. **Use in 3D CNN:** In activity classification, a 3D CNN uses 3D kernels ($k_x \times k_y \times k_t$). As the network gets deeper, the temporal receptive field grows, allowing the network to capture **long-range temporal dependencies** and complex motion patterns (like "running" or "waving") that span multiple frames, rather than just static appearance features.

解答 (中文): 含义: 在视频处理中, 感受野指的是输入视频体 (空间高/宽 $\times$ 时间帧) 中影响深层神经元激活的区域。它是**时空的 (spatio-temporal)**, 覆盖特定持续时间内的特定像素块。**在 3D CNN 中的使用:** 在活动分类中, 3D CNN 使用 3D 核。随着网络加深, 时间感受野增大, 使网络能够捕捉跨越多个帧的**长期时间依赖关系**和复杂的运动模式 (如 “跑步” 或 “挥手”), 而不仅仅是静态外观特征。

Section B: Long Question (50%)

Q11 Projection Matrix [15%]

Question Summary: (a) Parallel optical axes, C_2 at $[T_x, 0, 0]^T$ in C_1 's frame. Find (i) Rotation matrices, (ii) Essential matrix. (b) Non-parallel axes. $M_1 = K_1[I|0]$, $M_2 = K_2[R|T]$. Find (i) C_2 world location, (ii) Essential matrix.

Translation: (a) 光轴平行, C_2 在 C_1 坐标系中的位置为 $[T_x, 0, 0]^T$ 。求 (i) 旋转矩阵, (ii) 本质矩阵。 (b) 光轴不平行。 $M_1 = K_1[I|0]$, $M_2 = K_2[R|T]$ 。求 (i) C_2 的世界坐标位置, (ii) 关联两相机的本质矩阵。

Answer: (a) Parallel Axes:

(i) **Rotation Matrices:** Since axes are parallel and aligned, there is no relative rotation.

$$R_1 = I, \quad R_2 = I$$

(ii) **Essential Matrix (E):** $E = [t]_{\times} R$. Here, relative translation $t = [T_x, 0, 0]^T$ and $R = I$. The skew-symmetric matrix $[t]_{\times}$ is:

$$[t]_{\times} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -T_x \\ 0 & T_x & 0 \end{pmatrix}$$

$$\text{So, } E = [t]_{\times} I = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -T_x \\ 0 & T_x & 0 \end{pmatrix}.$$

(b) General Case:

(i) **Camera C_2 location:** The projection matrix is $M_2 = K_2[R|T]$. The camera center C satisfies $M_2 \tilde{C} = 0$.

$$RC + T = 0 \implies RC = -T \implies C = -R^T T$$

So, C_2 is located at $-R^T T$ in the world coordinate system.

(ii) **Essential Matrix (E):** The Essential Matrix relates normalized coordinates. Let $x_2 E x_1 = 0$. It is defined by the relative motion from C_1 to C_2 . Since C_1 is at the origin with identity rotation ($M_1 = K_1[I|0]$), the extrinsic parameters of C_2 (R, T) directly describe the relative motion. Thus, $E = [T]_{\times} R$.

解答 (中文): (a) 平行轴: (i) $R_1 = I, R_2 = I$ 。 (ii) 本质矩阵 $E = [t]_{\times} R = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -T_x \\ 0 & T_x & 0 \end{pmatrix}$ 。

(b) 一般情况: (i) 相机 C_2 的世界坐标位置为 $C = -R^T T$ 。 (ii) 本质矩阵为 $E = [T]_{\times} R$ 。

Q12 Stereo Vision [15%]

Question Summary: Left camera points straight. Right camera angled at 45 degrees. (a) Locations of epipoles? (b) Draw 5 epipolar lines on each image. (c) How to rectify? (d) How to estimate depth?

Translation: 左相机直指前方, 右相机成45度角。 (a) 极点在哪里? (b) 在两张图上画出至少5条极线。 (c) 描述如何校正图像对。 (d) 已知参数, 如何估计深度?

Answer: (a) Epipoles: The epipole is the image of the other camera's center.

- **Left Epipole (e_L):** The projection of the Right Camera Center onto the Left Image. Since the Right Camera is to the right of the Left Camera, e_L is located on the horizontal axis, to the right side (possibly at infinity if strictly parallel, but here cameras are converging, so it's a finite point).
- **Right Epipole (e_R):** The projection of the Left Camera Center onto the Right Image. Since the Right Camera is rotated 45 degrees looking leftwards towards the Left Camera, the Left Camera center will appear somewhere in the left-part of the Right Image.

(b) Epipolar Lines (Description):

- **Left Image:** The lines should all converge to the epipole e_L on the right side. They will look like a fan opening to the left.
- **Right Image:** The lines should all converge to the epipole e_R on the left side. They will look like a fan opening to the right.

(c) Rectification:

1. Rotate the Left Camera (virtually) and the Right Camera (virtually) so that both their optical axes are perpendicular to the baseline connecting the two optical centers.
2. Scale/warp the images so that the epipoles move to infinity.
3. This results in epipolar lines becoming horizontal and parallel (scanlines), meaning y -coordinates of matching points are identical.

(d) Depth Estimation: Given rectified images:

1. For a pixel (x_L, y) in the left image, search for the matching pixel (x_R, y) in the right image along the same scanline.
2. Calculate **disparity** $d = x_L - x_R$.
3. Calculate depth Z using the formula: $Z = \frac{f \cdot b}{d}$, where f is the focal length and b is the baseline distance between camera centers.

解答 (中文): (a) 极点: 左图像的极点在右侧 (右相机中心的投影)。右图像的极点在左侧 (左相机中心的投影)。(b) 极线: 左图: 向右侧极点汇聚的放射状线。右图: 向左侧极点汇聚的放射状线。(c) 校正: 旋转两个相机, 使其光轴都垂直于连接两个光心的基线。这将极点移至无穷远, 使极线变为水平平行线。(d) 深度估计: 在校正后的图像中计算视差 $d = x_L - x_R$ 。深度 $Z = \frac{f \cdot b}{d}$ 。

Q13 Image Transformation [10%]

Question: Poster plane $(X, Y, 0)$ to Image (u, v) . (a) Geometric transformation matrix? (b) Number of unknown parameters? (c) Min correspondence points? (d) What if more points? (e) Two invariant properties?

Translation: 海报平面 $(X, Y, 0)$ 到图像 (u, v) 。(a) 几何变换矩阵? (b) 未知参数数量? (c) 最少对应点数? (d) 点多于最少点数会怎样? (e) 两个不变性质?

Answer: (a) Transformation: This is a planar homography (Projective Transformation).

$$s \begin{pmatrix} u \\ v \\ 1 \end{pmatrix} = H \begin{pmatrix} X \\ Y \\ 1 \end{pmatrix} = \begin{pmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{pmatrix} \begin{pmatrix} X \\ Y \\ 1 \end{pmatrix}$$

(Note: Since $Z=0$, the 3×4 projection matrix reduces to a 3×3 homography).

(b) Unknowns: The matrix has 9 entries, but is defined up to a scale factor. Thus, there are **8** degrees of freedom (unknowns).

(c) Minimum points: Each point correspondence provides 2 constraints (x and y). $8 \div 2 = 4$. So, **4 points** are required.

(d) Over-determined: If more points are available, the system is over-determined. We cannot find an exact solution for all points due to noise. We use optimization methods (like Singular Value Decomposition / DLT) to find the solution that **minimizes the re-projection error** (Least Squares solution).

(e) Invariants: 1. **Collinearity:** Straight lines remain straight lines. 2. **Cross-ratio:** The cross-ratio of 4 collinear points is preserved. (Note: Parallelism is NOT preserved).

解答 (中文): (a) 单应性矩阵 (Homography), 3×3 矩阵。 (b) 8个未知参数 (9个元素减去1个尺度因子)。 (c) 最少需要 4 个对应点。 (d) 系统变为超定 (Over-determined)。使用最小二乘法 (如 DLT/SVD) 求解, 以最小化重投影误差。 (e) 共线性 (Collinearity) 和 交比 (Cross-ratio)。

Q14 Optical Flow [10%]

Question: $I(x+u, y+v, t+1) = aI(x, y, t) + b$. Show linear system for (u, v, a, b) . Min window size?

Translation: $I(x+u, y+v, t+1) = aI(x, y, t) + b$. 展示 (u, v, a, b) 的线性方程组。最小窗口大小?

Answer: Derivation: 1. First order Taylor expansion of LHS: $I(x+u, y+v, t+1) \approx I(x, y, t+1) + I_x u + I_y v$ 2. Substitute into equation: $I(x, y, t+1) + I_x u + I_y v = aI(x, y, t) + b$ 3. Rearrange to put unknowns (u, v, a, b) on LHS. Note that this is non-linear due to a . However, usually, we assume $a \approx 1$ for linearization or treat the change differently. The question asks for a linear system. Let's rewrite: $I_x u + I_y v - I(x, y, t)a - b = -I(x, y, t+1)$ This is a linear equation in 4 unknowns: u, v, a, b .

$$\begin{bmatrix} I_x & I_y & -I & -1 \end{bmatrix} \begin{bmatrix} u \\ v \\ a \\ b \end{bmatrix} = -I_t - I$$

(Note: Often approximated as $I_x u + I_y v + I_t \approx (a-1)I + b$). Let's stick to the direct form derived: $I_x u + I_y v - I_{curr} a - b = -I_{next}$.

Minimum Window Size: We have 4 unknowns (u, v, a, b) but only 1 equation per pixel. To solve it, we need at least 4 equations. Thus, we need a window of at least **4 pixels** (e.g., 2×2 window) assuming the parameters are constant within the window.

解答 (中文): 推导: 对左边进行泰勒展开: $I(x, y, t+1) + I_x u + I_y v \approx aI(x, y, t) + b$ 。整理得到线性方程: $I_x u + I_y v - I(x, y, t) \cdot a - 1 \cdot b = -I(x, y, t+1)$ 。这是一个关于 u, v, a, b 的线性方程。

最小窗口: 因为有 4 个未知数, 每个像素只提供 1 个方程, 所以至少需要 **4 个像素** (例如 2×2 的窗口) 来求解。